Exp No:20 K-Means Clustering.

Program:

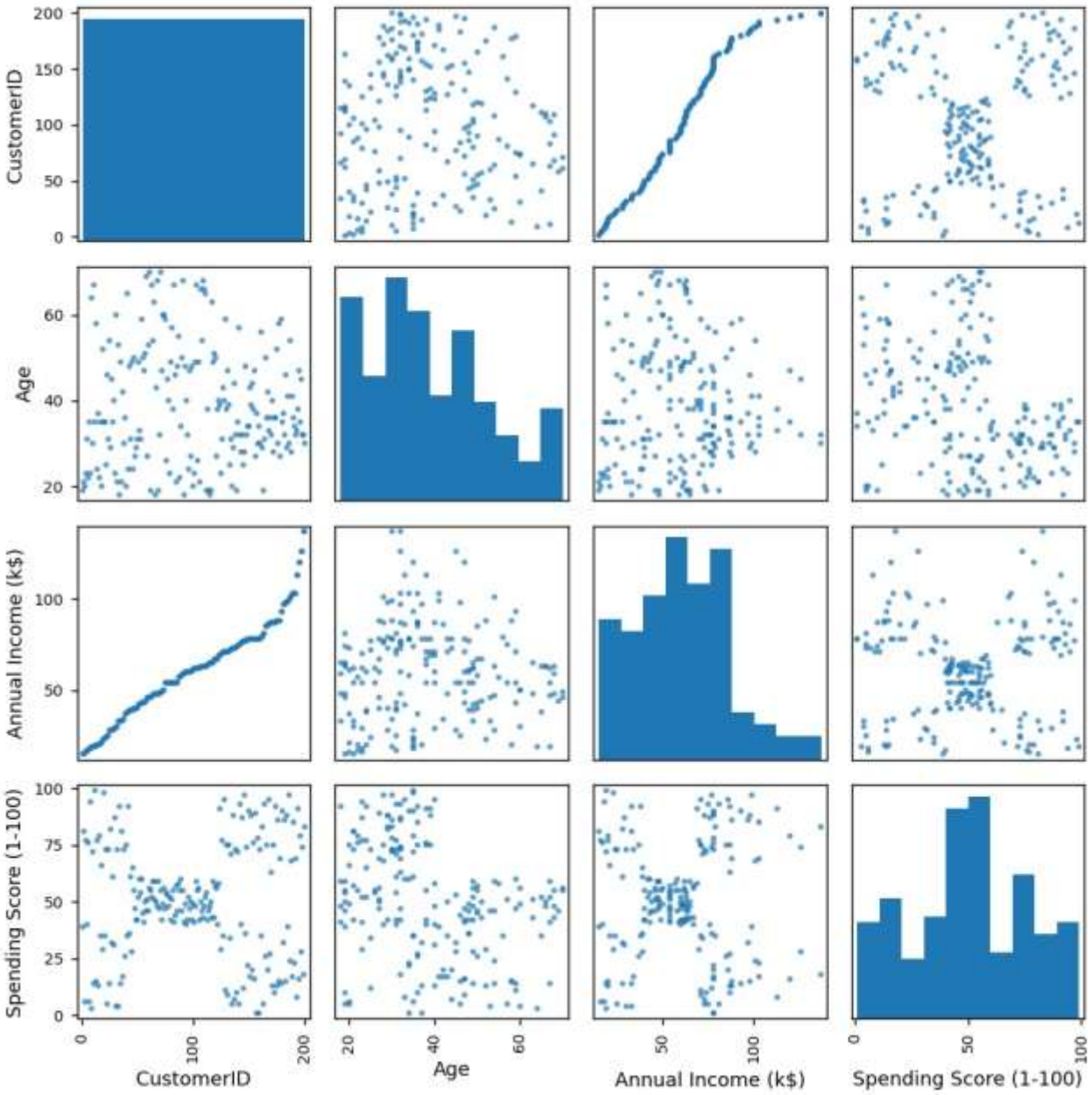
```
import numpy as np
import pandas as pd
import
matplotlib.pyplot as
plt import seaborn as
sns %matplotlib inline
df=pd.read_csv('Mall_C
ustomers.csv')
df.info() df.head()
sns.pairplot(df)
features=df.iloc[:,[3,
4]].values from
sklearn.cluster import
KMeans
model=KMeans(n_cluster
s=5)
model.fit(features)
Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
Final.head() sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8)
\ .map(plt.scatter,"Annual Income (k$)",
"Spending Score (1-100)") \ .add_legend()
plt.show()
features_el=df.iloc[:,[2,3,4]].values from
sklearn.cluster import KMeans wcss=[] for
i in range(1,10):
model=KMeans(n_clusters=i)
model.fit(features_el)
wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200
entries, 0 to 199
Data columns (total
5columns):
0 CustomerID              200 non-null    int64
1 Gender                  200 non-null    object
2 Age                     200 non-null    int64
3 Annual Income (k$)      200 non-null    int64 4   Spending Score
(1-100)  200 non-null    int64 dtypes: int64(4), object(1) memory
usage: 7.9+ KB
```

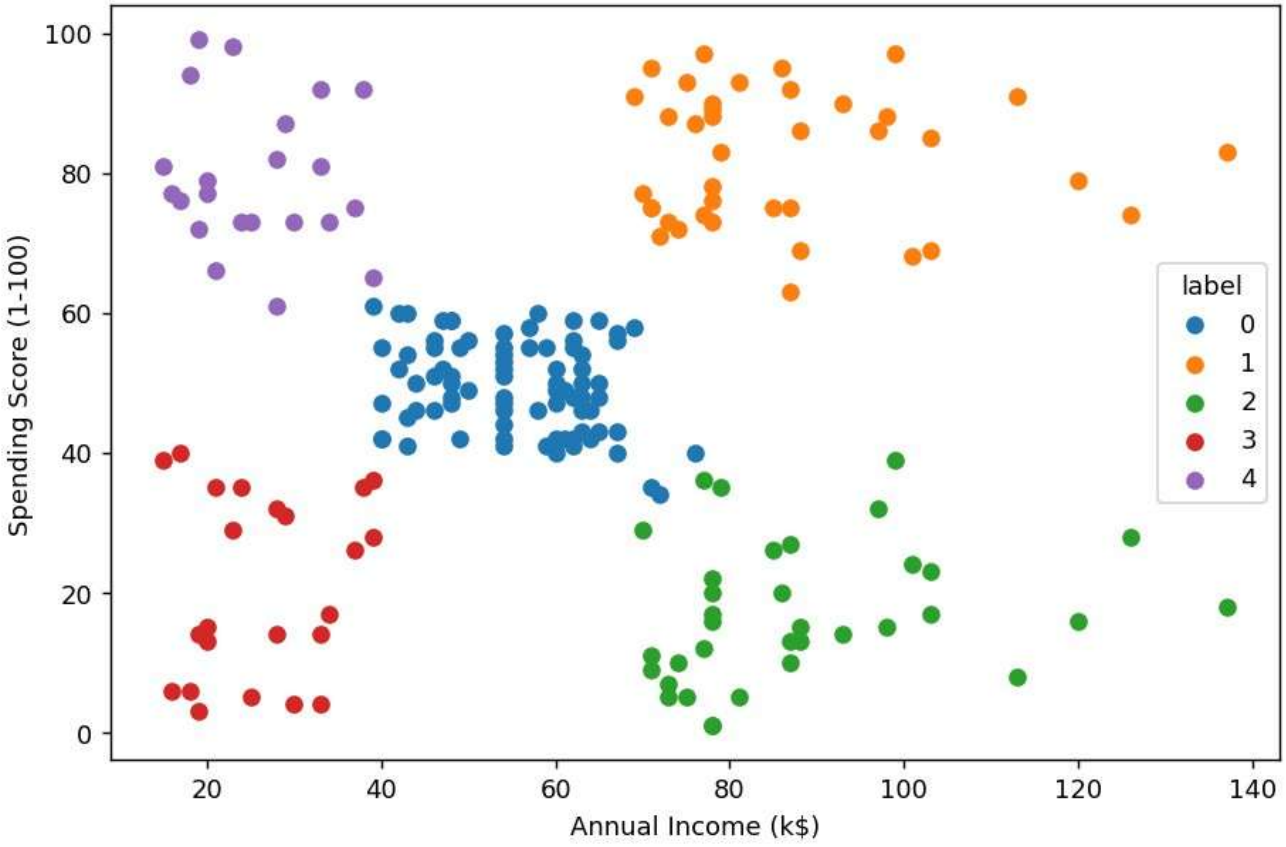
```
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-
100)
```

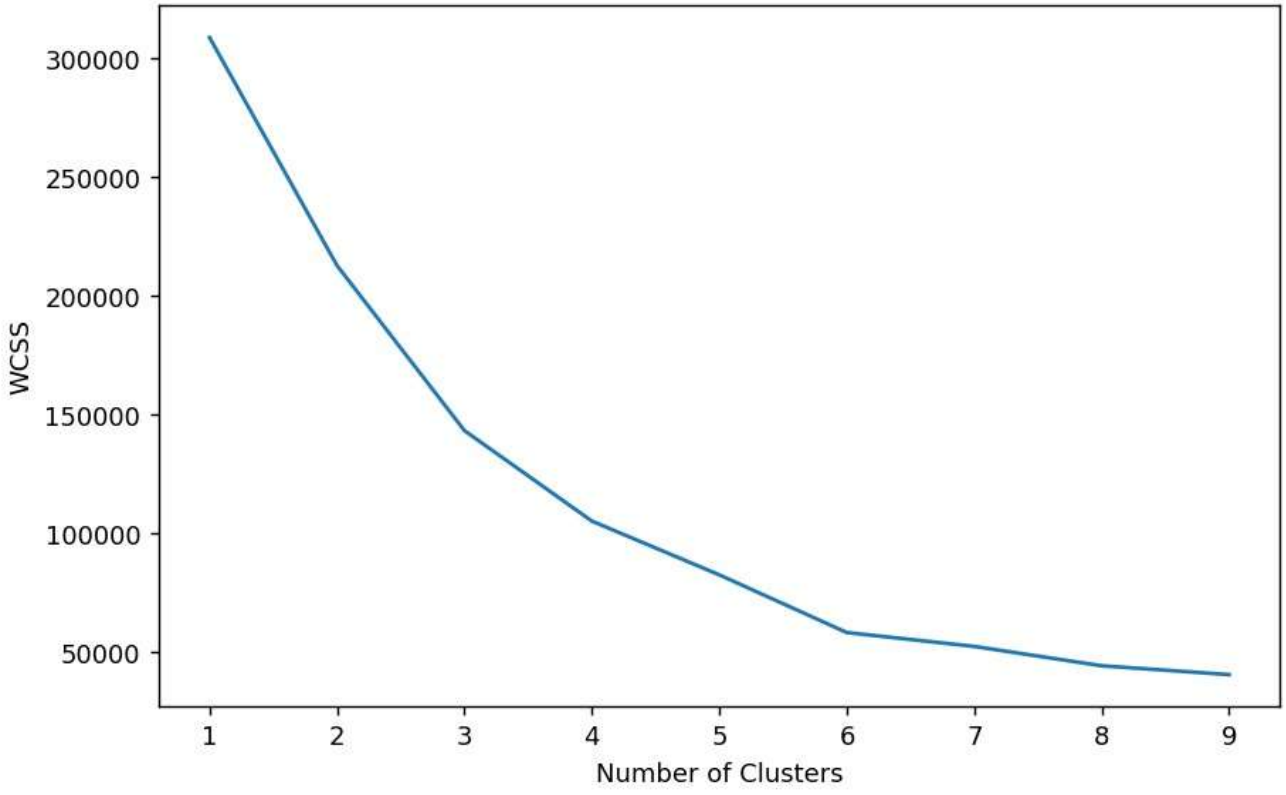
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40


CustomerID
Age
Annual Income (k$)
Spending Score (1-100)
0
50
100
150
200
20
40
60
25
75

Output

	Annual Income (k$)	Spending Score (1-100)	label
0	15	39	3
1	15	81	4
2	16	6	3
3	16	77	4
4	17	40	3


Spending Score (1-100)
Annual Income (k$)
label
0
1
2
3
4
100
80
60
40
20
0
20
40
60
80
100
120
140


WCSS
Number of Clusters
300000
250000
200000
150000
100000
50000
1
2
3
4
5
6
7
8
9